

AnchorBench: A Theory-Grounded Benchmark for Anchoring in Large Language Models

Anonymous authors

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Abstract

1 Anchoring—the tendency for numeric estimates to shift toward contextually
 2 salient reference points—is widely documented in both human cognition
 3 and large language models (LLMs), yet existing LLM evaluations treat
 4 anchoring monolithically: they test a single delivery mechanism, conflate
 5 sensitivity to irrelevant numbers with rational use of informative context,
 6 and lack grounding in the distinct psychological paradigms that produce
 7 anchoring in humans. We introduce ANCHORBENCH, a theory-grounded
 8 benchmark that maps four established human anchoring paradigms—classic
 9 external anchoring, self-generated anchoring, numeric priming, and ecological
 10 anchoring—to five LLM interface channels (prompt-embedded, conversation
 11 history, in-context learning, retrieval-augmented generation, and tool-call
 12 outputs). Crucially, the benchmark introduces an explicit *anchor-relevance*
 13 *axis* that distinguishes transparently irrelevant anchors from plausibly
 14 informative ones, enabling measurement of both raw susceptibility and
 15 relevance discrimination—a reliability dimension invisible to standard accuracy
 16 metrics. Pilot studies on all five suites (external, history, ICL, RAG, and tool)
 17 across four models validate the design using a unified evaluation framework.
 18 The Unified Anchor Influence (UAI) metric—which measures the fraction of
 19 available shift toward the anchor that is realized—enables direct comparison
 20 across suites with different anchor-delivery mechanisms. Results show that
 21 anchoring susceptibility is paradigm-dependent: model rankings on both task
 22 accuracy and discrimination reverse across suites, and no model is consistently
 23 best or worst. Plausible anchors produce stronger influence than irrelevant
 24 ones in 18 of 20 model-suite combinations, but the magnitude and consistency
 25 of discrimination vary substantially, confirming the value of multi-paradigm
 26 evaluation.

27 1 Introduction

28 When LLMs are deployed for numeric estimation in high-stakes domains—medical
 29 triage scores, financial risk indices, operational efficiency ratings—their outputs
 30 must be robust to irrelevant contextual numbers. Anchoring, the systematic
 31 tendency to shift estimates toward salient reference points (Tversky & Kahneman,
 32 1974), is among the most robust biases documented in human judgment
 33 (Kahneman, 2011; Chapman & Johnson, 1999). Recent work has confirmed
 34 that LLMs exhibit analogous behavior: model estimates shift toward numbers
 35 embedded in prompts, conversation history, and retrieved context (Jones &
 36 Steinhardt, 2022; Hagendorff et al., 2023; Echterhoff et al., 2024; Huang
 37 et al., 2025; Lou & Sun, 2025). Unlike generic accuracy degradation, anchoring
 38 is particularly insidious because biased outputs can appear confident and
 39 well-reasoned, making the failure mode difficult to detect without controlled
 40 evaluation.

41 However, existing LLM anchoring evaluations are fragmented and insufficient
 42 for systematic benchmarking. Most studies test only prompt-embedded anchors,
 43 ignoring the multiple interfaces through which anchors reach real LLM systems—conversation
 44 history, few-shot demonstrations, retrieved documents, and tool outputs. More
 fundamentally, current evaluations treat *all* numeric sensitivity as uniformly
 undesirable, conflating irrational susceptibility to irrelevant numbers with
 potentially rational use of informative context. A model

that appropriately adjusts its estimate given a credible prior is behaving differently from one that shifts toward an arbitrary case ID—yet existing metrics cannot distinguish these behaviors. Finally, few studies ground their designs in the cognitive science literature, which identifies multiple distinct anchoring mechanisms (Epley & Gilovich, 2001; Mussweiler & Strack, 1999; Chapman & Johnson, 1999) that may map onto different LLM failure modes. In short, prior work establishes *that* LLMs can anchor but leaves open *when* and *through which interface* anchoring arises, and whether sensitivity tracks the normative relevance of the anchor.

We introduce ANCHORBENCH, a benchmark that addresses these limitations through two key design choices. First, we map four established human anchoring paradigms—classic external anchoring (Tversky & Kahneman, 1974), self-generated anchoring via insufficient adjustment (Epley & Gilovich, 2001), numeric priming through selective accessibility (Mussweiler & Strack, 1999), and ecological anchoring in realistic information environments—to five LLM interface channels (prompt-embedded, conversation history, in-context learning, retrieval-augmented generation, and tool-call outputs). Second, we introduce an explicit *anchor-relevance axis* with four levels (none, irrelevant, plausible, informative) that operationalizes the normative status of each anchor. This axis enables the benchmark to measure not just how much a model’s estimates shift, but whether the model discriminates between anchors it should ignore and anchors that carry genuine evidential weight.

Our contributions are:

1. **Theory-grounded benchmark design.** A systematic mapping from four human anchoring paradigms to five LLM interface channels, with an anchor-relevance axis that separates susceptibility from discrimination (§3).
2. **Evidence-first task construction.** Each benchmark item presents structured numeric evidence from which a gold answer (y_{evidence}^*) is derivable, ensuring that anchoring is measured against a well-defined rational baseline (§3.3).
3. **Unified evaluation framework.** A single set of metrics—Unified Anchor Influence (UAI), Toward-Anchor Rate (TAR), and discrimination gap (Disc_Δ)—that enables direct comparison across suites with different anchor-delivery mechanisms, replacing suite-specific metrics with a coherent framework (§4).
4. **Five-suite pilot validation.** Pilot studies on the external suite (90 items, 4 models), the history suite (60 items, two-stage self-generated protocol), the ICL suite (180 items, metadata priming in few-shot demonstrations), the RAG suite (180 items, three-document corpus design), and the tool suite (180 items, structured tool outputs via chat template) demonstrate that anchoring susceptibility is paradigm-dependent: model rankings on both accuracy and discrimination reverse across suites, and plausible anchors produce stronger influence than irrelevant ones in 18 of 20 model–suite combinations (§5).

We release the benchmark, generation pipeline, and evaluation code to support reproducible research on LLM anchoring.

2 Related work

Human anchoring theory. Anchoring was first identified by Tversky & Kahneman (1974) as one of three judgment heuristics, demonstrated through the classic experiment in which randomly generated numbers influenced estimates of factual quantities. Subsequent research revealed that anchoring is not a unitary phenomenon. Epley & Gilovich (2001) distinguished *self-generated* anchoring, where individuals start from a self-produced value and adjust insufficiently, from *experimenter-provided* anchoring, which operates through different cognitive pathways. Mussweiler & Strack (1999) proposed the *selective accessibility* model, in which anchors prime anchor-consistent knowledge via hypothesis-consistent testing, operating even when the anchor is transparently uninformative. Chapman & Johnson (1999) reviewed anchoring in applied judgment, identifying conditions under which it is amplified (uncertainty, cognitive load) or attenuated (expertise, incentives). A key lesson from this literature is

that different paradigms produce anchoring through distinct mechanisms, and that rational agents should respond differently to informative versus uninformative reference points—a distinction that motivates the relevance axis in our benchmark.

Anchoring and cognitive biases in LLMs. Several studies have documented anchoring-like behavior in LLMs. Jones & Steinhardt (2022) mapped human cognitive biases to LLM failure modes, proposing a taxonomy that includes anchoring. Hagendorff et al. (2023) found that GPT-family models exhibited anchoring and framing effects that diminished with instruction tuning. Echterhoff et al. (2024) systematically tested 14 cognitive biases across multiple LLMs. Itzhak et al. (2023) distinguished instructed and default anchoring, showing that models vary in susceptibility. More recently, Lou & Sun (2025) tested anchoring across GPT-4 and Gemini with multiple mitigation strategies, Huang et al. (2025) studied anchoring mechanisms and mitigations using synthetic anchors, Takenami et al. (2025) demonstrated anchoring effects in LLM-driven price negotiation simulations, and Valencia-Clavijo et al. (2025) combined behavioral analysis with Shapley-value attribution to measure anchor influence on output distributions. The `tum-nlp` cognitive bias dataset (Malberg et al., 2025) provides control/treatment pairs for multiple biases but uses ordinal Likert scales rather than continuous numeric estimates. While these studies collectively establish that LLMs exhibit anchoring, they share common limitations: most test only one or two delivery mechanisms (typically prompt-embedded), none introduces a relevance-control axis, and none provides a systematic mapping to the cognitive science paradigms that distinguish anchoring mechanisms.

Bias benchmarks and evaluator bias. Beyond anchoring-specific work, the broader LLM evaluation literature highlights the importance of controlled bias measurement. Koo et al. (2024) showed that LLMs acting as evaluators exhibit systematic cognitive biases, suggesting that anchoring effects extend beyond estimation tasks to evaluation settings. Bias benchmarks for social biases have established the value of controlled, paired evaluation designs, but no analogous theory-grounded benchmark exists for cognitive biases in numeric estimation. Our work fills this gap by providing a benchmark specifically designed to separate susceptibility from relevance discrimination—a distinction that, to our knowledge, no existing LLM anchoring evaluation operationalizes.

3 Benchmark design

We present ANCHORBENCH, a benchmark for measuring anchoring in LLM numeric estimation. Unlike prior work that tests a single anchor-delivery mechanism and treats all numeric sensitivity as equally undesirable, ANCHORBENCH is organized around two orthogonal axes: the *source* through which anchors reach the model (mapped to human anchoring paradigms) and the *relevance* of the anchor (determining whether sensitivity is rational).

3.1 Theory-to-benchmark mapping

The cognitive science literature identifies multiple mechanisms that produce anchoring in human judgment. We map four such paradigms to five LLM interface channels, creating benchmark suites that test distinct failure modes (Table 1).

3.2 Anchor-relevance axis

The benchmark’s key methodological innovation is separating anchor *source* from anchor *relevance*. Not all numeric sensitivity is irrational: a model that adjusts its estimate toward a validated prior is behaving differently from one that shifts toward an arbitrary batch number. We define four relevance levels:

- **None** (control): No anchor-bearing numeric cue appears beyond the task evidence. The model should rely on evidence alone.

Paradigm	Human mechanism	Suite	LLM interface
P1: External	Classic experimenter-provided anchoring (Tversky & Kahneman, 1974)	External	Numeric cue embedded in the prompt text
P2: Self-gen.	Insufficient adjustment from self-generated starting point (Epley & Gilovich, 2001)	History	Model’s own prior estimate in conversation history
P3: Priming	Semantic / numeric priming via selective accessibility (Mussweiler & Strack, 1999)	ICL	Irrelevant metadata numbers in few-shot demonstrations
P4: Ecological	Anchoring in realistic information environments (Chapman & Johnson, 1999)	RAG, Tool	Numeric content in retrieved documents or structured tool outputs

Table 1: Mapping from human anchoring paradigms to ANCHORBENCH suites. Each suite tests a distinct source through which an anchor can influence LLM estimates.

- **Irrelevant:** A transparently arbitrary number with no evidential relation to the target (e.g., a case ID, batch number, or sequential request index). A well-calibrated model should exhibit near-zero sensitivity.
- **Plausible:** A weakly credible, plausibly related numeric cue (e.g., “a recent industry report suggested...”), but not validated enough to override the provided evidence. Limited sensitivity may be defensible, but the model should remain primarily evidence-driven.
- **Informative:** A genuinely task-relevant signal (e.g., a validated prior-quarter measurement). Some sensitivity is desirable if the signal is genuinely predictive.

The core benchmark conditions are `control` + `irrelevant_low` + `irrelevant_high` + `plausible_low` + `plausible_high` (5 prompt views per item). The `informative` level is reserved for optional validation experiments. This design enables the benchmark to measure both *susceptibility* (overall shift toward anchors) and *discrimination* (whether sensitivity depends on relevance).

3.3 Evidence-first task construction

Every item presents the model with a numeric estimation task in which structured evidence is the primary input. The construction ensures that a deterministic gold answer is derivable from the evidence alone, independent of any anchor.

Latent signal. For each item, a latent parameter θ is drawn uniformly from $[30, 70]$ (seed-controlled). Five numeric ratings are sampled i.i.d. from $\mathcal{N}(\theta, \sigma^2)$ with $\sigma = 8$, clamped to

[0, 100] and rounded to integers. The evidence is purely informational and correctly centered on θ ; no anchor enters the evidence block.

Gold answer. The evaluation target is $y_{\text{evidence}}^* = \text{round}(\bar{e})$, the rounded mean of the evidence values actually presented to the model. This ensures that a model performing correct evidence aggregation receives full credit, regardless of sampling noise between θ and the evidence mean. The latent θ is retained as y_{θ}^* for understanding the data generation process but is not the primary evaluation target.

Anchor construction. Anchors are placed symmetrically around θ using stratified offsets $\delta \in \{15, 25, 40\}$: $a_{\text{low}} = \max(0, \theta - \delta)$, $a_{\text{high}} = \min(100, \theta + \delta)$. Item generation iterates over offset levels to ensure balanced coverage, enabling analysis of whether anchoring effects scale with anchor–target distance. (For the History suite, the anchor is the model’s own Stage 1 answer rather than an experimenter-defined value; see §4.)

Domains. Each item asks the model to estimate a domain-specific index on a 0–100 scale. Six domains span distinct estimation contexts to avoid domain-specific confounds: pricing / willingness-to-pay, operations / time estimation, transportation / logistics reliability, resource consumption / efficiency, market / demographics adoption, and legal / policy compliance. Each domain defines five evidence-label families, multiple scenario templates, question templates requesting a 0–100 integer, and relevance-specific anchor-preamble templates.

Difficulty axis. Items vary in estimation difficulty:

- **Easy:** five concordant ratings with $\sigma = 8$; the evidence mean is close to θ and the answer is straightforward to derive.
- **Hard:** three ratings (two marked as unavailable) with $\sigma = 15$ and one conflicting value; the evidence is sparse and ambiguous.

The human anchoring literature predicts that anchoring is stronger under uncertainty (Chapman & Johnson, 1999); this axis enables testing that prediction in LLMs.

3.4 Benchmark suites

Each suite instantiates a specific human anchoring paradigm through a realistic LLM interface. For each suite we state: (1) what paradigm it instantiates, (2) why the interface is realistic, and (3) what a rational model should do under the relevance axis.

External suite (Paradigm P1: classic external anchoring). The anchor appears as a sentence in the prompt text, replicating the classic paradigm from Tversky & Kahneman (1974) in which an experimenter provides a numeric reference point before eliciting a judgment. This is the most direct LLM interface for external anchoring: users, system prompts, or upstream pipelines routinely embed numeric cues in prompt context. *Irrelevant condition:* “Note: This is assessment case #a in the current evaluation batch.”—a transparently arbitrary identifier. *Plausible condition:* “A recent industry report suggested the [metric] index might be around a.”—a weakly credible estimate from an unspecified source. *Rational behavior:* A well-calibrated model should show near-zero shift for the irrelevant case ID and at most limited shift for the plausible report, remaining primarily evidence-driven in both cases. Control prompts omit the anchor sentence entirely. A strict pairing invariant is enforced: anchored and control prompts differ only in the anchor-bearing sentence.

History suite (Paradigm P2: self-generated anchoring). The anchor originates from the model’s own prior estimate, instantiating Epley & Gilovich (2001)’s self-generated anchoring paradigm, in which insufficient adjustment from a self-produced starting point biases the final judgment. In deployed chat systems, a model’s own prior output—cached in the conversation turn—can anchor subsequent estimates. We implement a genuine two-stage protocol with two condition families: *Plausible (same-case):* In Stage 1, the model estimates from a partial-evidence subset (2 of 5 ratings selected to bias the initial estimate low or

high). In Stage 2, full evidence is revealed in the same conversation and the model re-estimates. The Stage 1 answer becomes a self-generated anchor; insufficient adjustment toward the evidence-supported answer constitutes anchoring. *Irrelevant (different-case)*: In Stage 1, the model answers a warm-up item from the same domain but a different case (with evidence designed to produce a low or high estimate). In Stage 2, the target item is presented as explicitly new, with the warm-up answer retained in conversation history. Any influence of the irrelevant warm-up answer on the target estimate constitutes anchoring. *Control*: A single-stage prompt with full evidence and no prior history, identical to the external-suite control. *Rational behavior*: Under plausible conditions, a rational model should adjust nearly completely from its Stage 1 estimate toward the full-evidence answer. Under irrelevant conditions, it should produce estimates indistinguishable from control, since the warm-up answer pertains to a different case.

ICL suite (Paradigm P3: numeric priming). Three few-shot demonstrations precede the target question. This suite instantiates Mussweiler & Strack (1999)’s selective accessibility model: irrelevant numbers in the demonstration context may prime anchor-consistent retrieval of numeric knowledge, even when the numbers have no evidential relation to the target. Few-shot prompting is a standard LLM interface, and demonstration context routinely contains incidental metadata (example IDs, dataset indices, timestamps) that could function as numeric primes. In the current design, demo answers are kept neutral to isolate the priming effect of metadata numbers from ICL distribution-matching. *Irrelevant condition*: Demo headers include case IDs and batch numbers near the anchor value: “*Example 1 [Case #a, Batch a]: ...*” *Rational behavior*: Metadata identifiers carry no information about the target quantity; a well-calibrated model should show no sensitivity to these numbers.

RAG suite (Paradigm P4: ecological anchoring). The anchor appears within a retrieved document, reflecting a realistic deployment scenario in which a retrieval-augmented system surfaces documents containing task-related (or task-unrelated) numeric content. This is not a direct instantiation of a classic psychology paradigm but rather an ecological extension: in deployed RAG pipelines, models routinely encounter numeric values in retrieved passages whose relevance varies from incidental to highly informative. Each item has a frozen three-document mini-corpus: (i) a *core* document providing genuinely relevant domain context (no anchor), (ii) a *filler* document with neutral methodological background (no anchor), and (iii) an *anchor-slot* document whose content varies across conditions. Retrieval is deterministic: documents are presented in fixed order [core, anchor-slot, filler], ensuring that the only variation across matched conditions is the anchor-slot document content. *Control condition*: The anchor-slot document is a neutral administrative note: “*No external benchmark data was located for this assessment case.*” *Irrelevant condition*: The anchor-slot document states: “*Administrative note: document batch reference #a. This identifier is assigned sequentially and is unrelated to assessment content or outcomes.*” *Plausible condition*: The anchor-slot document states: “*A recent survey of comparable organizations reported a [metric] index of approximately a.*” *Rational behavior*: Administrative metadata should be ignored entirely; a plausible survey result may warrant limited attention but should not override direct evidence.

Tool suite (Paradigm P4: ecological anchoring). The anchor is returned within a structured tool output rendered using the model’s native tool-calling chat template (with `role: "tool"` messages), making the framing structurally distinct from prompt-embedded text. In deployed agentic systems, LLMs routinely receive structured JSON data from external tools—APIs, database lookups, monitoring dashboards—whose outputs contain both informative fields and incidental metadata (request IDs, batch indices, sequence numbers). Two deterministic tools are available: `get_evidence_summary` (computes summary statistics over the evidence ratings) and `check_external_reference` (looks up external benchmark data for a domain). The prompt includes a simulated tool-call exchange: the assistant issues both tool calls, and the tool responses appear as structured `role: "tool"` messages. The `get_evidence_summary` output is identical across all conditions; only `check_external_reference` varies: *Control condition*: `{"status": "no_data", "note": "No external reference available..."}`. *Irrele-*

Condition	Relevance	Anchor
control	none	—
irrelevant_low	irrelevant	$\theta - \delta$
irrelevant_high	irrelevant	$\theta + \delta$
plausible_low	plausible	$\theta - \delta$
plausible_high	plausible	$\theta + \delta$

Table 2: Core benchmark conditions per item. Each item produces five PROMPTVIEWS that share identical evidence and question; only the anchor-bearing component varies. For the External, ICL, RAG, and Tool suites, anchors are experimenter-provided at $\theta \pm \delta$ with $\delta \in \{15, 25, 40\}$. For the History suite, anchors are self-generated: the model’s own Stage 1 answer (plausible) or a warm-up answer from a different case (irrelevant). The **informative** level is available as an optional validation condition.

268 *vant condition*: {"request_id": a , "status": "no_data", "note": "Request ID
269 is a sequential processing identifier..."—the anchor appears only as a metadata
270 field explicitly labeled as unrelated to content. *Plausible condition*: {"reference_value":
271 a , "source": "industry_survey_2025", "confidence": "low"}—the anchor is a
272 nominally informative but low-confidence benchmark value. *Rational behavior*: A request ID
273 is pure metadata and should be ignored; a low-confidence benchmark value from an unver-
274 ified survey may warrant limited attention but should not override direct evidence. An
275 ecological validation subset additionally tests a small number of items (~ 50) through gen-
276 uine multi-turn function-calling interaction, where the model generates tool calls, receives
277 executed results, and produces a final answer, validating that the injected results generalize
278 to authentic agentic behavior.

279 3.5 Benchmark conditions and scale

280 Table 2 summarizes the benchmark conditions.

281 **Scale.** The external-suite pilot uses $90 \text{ items} \times 5 \text{ conditions} = 450$ prompts across 6 do-
282 mains. The history-suite pilot uses $60 \text{ items} \times 5 \text{ conditions} = 300$ prompts across the
283 same 6 domains, with each non-control condition requiring a two-stage conversation. The
284 RAG-suite pilot uses $180 \text{ items} \times 5 \text{ conditions} = 900$ prompts across 6 domains, with a
285 frozen three-document corpus per item. The tool-suite pilot uses $180 \text{ items} \times 5 \text{ conditions}$
286 $= 900$ prompts across 6 domains, with structured tool outputs rendered via the model’s
287 chat template. The ICL-suite pilot uses $180 \text{ items} \times 5 \text{ conditions} = 900$ prompts across
288 6 domains, with three few-shot demonstrations whose metadata headers carry the anchor.
289 The full benchmark targets $5 \text{ suites} \times 6 \text{ domains} \times 20 \text{ items} \times 2 \text{ difficulty levels} = 1,200$
290 ITEMSPECS, each producing 5 PROMPTVIEWS, for 6,000 total prompts. All generation is
291 seed-controlled (seed = 42) and deterministic.

292 3.6 Prompt structure and validation

293 All prompts end with: “Return only a single integer 0–100 on the last line.” Each
294 PROMPTVIEW stores the full prompt text, structured prompt components, anchor string
295 and character-offset span (for mechanistic analyses), and a prompt hash for integrity checks.

296 Deterministic validators verify: (i) $\theta \in [30, 70]$, $y_{\text{evidence}}^* = \text{round}(\bar{e})$, anchors in $[0, 100]$,
297 offset $\delta \in \{15, 25, 40\}$; (ii) evidence ratings are present, in $[0, 100]$, and consistent with
298 domain labels; (iii) control and anchored prompts differ by at most the anchor-bearing
299 component; (iv) RAG document assertions and tool-call schemas are valid; (v) all prompts
300 contain the answer-format instruction. All five pilot datasets (external, history, ICL, RAG,
301 tool) pass all validators with zero errors. The history-suite validator additionally checks that
302 partial-evidence subsets bracket the gold answer and that warm-up cases are drawn from
303 the same domain as the target. The RAG-suite validator verifies that the three-document
304 corpus is present, that anchor-slot documents contain the correct anchor value, and that
305 retrieved document IDs are consistent across conditions. The tool-suite validator checks

that anchor values appear in the correct tool-output JSON field (`request_id` for irrelevant, `reference_value` for plausible), that `get_evidence_summary` output is identical across all conditions, and that prompt hashes are unique per condition. The ICL-suite validator checks that all five conditions share identical target evidence, scenario, question, demo evidence, and demo answers; only the demo header metadata differs across matched conditions.

4 Evaluation framework

All suites share a single evaluation framework. The unified metrics enable direct cross-suite comparison of anchoring susceptibility, regardless of how anchors reach the model.

Task accuracy (control-only). We report mean absolute error $\text{MAE}_{\text{ctrl}} = \frac{1}{n} \sum_i |y_{\text{ctrl},i} - y_{\text{evidence},i}^*|$ and accuracy-within- ± 10 $\text{Acc}_{10} = \frac{1}{n} \sum_i \mathbf{1}[|y_{\text{ctrl},i} - y_{\text{evidence},i}^*| \leq 10]$, computed exclusively on **control**-condition responses. Restricting to control ensures that task accuracy measures pure estimation ability without any anchor influence.

Unified Anchor Influence (UAI). For each item i under anchored condition r with anchor value $a_{i,r}$, we define:

$$\text{UAI}_{i,r} = \frac{y_{\text{anchor},i,r} - y_{\text{ctrl},i}}{a_{i,r} - y_{\text{ctrl},i}}, \quad (1)$$

where $y_{\text{anchor},i,r}$ is the model’s response under the anchored condition and $y_{\text{ctrl},i}$ is its matched control response for the same item. $\text{UAI} = 1$ indicates that the model’s answer shifts by the full anchor–control gap (complete susceptibility); $\text{UAI} = 0$ indicates no anchoring. Items where $|a_{i,r} - y_{\text{ctrl},i}| < \varepsilon$ are excluded ($\varepsilon = 3$) to avoid division by small denominators. Separate means are computed for irrelevant (UAI_{irr}) and plausible ($\text{UAI}_{\text{plaus}}$) conditions, each averaging over both low- and high-anchor items.

The UAI denominates by the anchor–control distance rather than the anchor–gold distance used in some prior work. This choice makes the metric interpretable as the fraction of the model’s available shift (from its own baseline toward the anchor) that is actually realized, and avoids conflating anchoring magnitude with task-accuracy errors in the control condition.

Anchor-value mapping. For the External, ICL, RAG, and Tool suites, $a_{i,r}$ is the experimenter-defined anchor value ($\theta \pm \delta$, where $\delta \in \{15, 25, 40\}$). For the History suite, $a_{i,r}$ is the model’s own Stage 1 answer, which serves as the self-generated anchor. This mapping unifies all five paradigms under a single formula while preserving the paradigm-specific semantics of what constitutes the anchor.

Toward-Anchor Rate (TAR). The binary toward-anchor rate measures the fraction of items where the model’s response moves in the direction of the anchor relative to its control baseline:

$$\text{TAR}_{i,r} = \mathbf{1}[(y_{\text{anchor},i,r} - y_{\text{ctrl},i}) \cdot (a_{i,r} - y_{\text{ctrl},i}) > 0]. \quad (2)$$

$\text{TAR} = 0.50$ indicates chance-level directional influence; $\text{TAR} > 0.50$ indicates systematic shift toward anchors. Unlike UAI, TAR does not require epsilon filtering and captures even small directional effects.

Discrimination gap. The discrimination gap quantifies whether a model differentiates between irrelevant and plausible anchors:

$$\text{Disc}_{\Delta} = \text{UAI}_{\text{plaus}} - \text{UAI}_{\text{irr}}. \quad (3)$$

$\text{Disc}_{\Delta} > 0$ indicates greater sensitivity to plausible than irrelevant anchors (the desired pattern); $\text{Disc}_{\Delta} \approx 0$ indicates indiscriminate anchoring; $\text{Disc}_{\Delta} < 0$ indicates greater sensitivity to irrelevant anchors—the worst-case failure mode. Unlike a ratio-based formulation, the difference is well-defined even when $\text{UAI}_{\text{plaus}} \approx 0$.

History-suite secondary metrics. The History suite’s two-stage protocol additionally supports two adjustment-specific metrics reported in Appendix B: the *Adjustment Completion Ratio* ($ACR = (y^{(2)} - y^{(1)}) / (y_{\text{evidence}}^* - y^{(1)})$), measuring how completely the model corrects from Stage 1 to Stage 2, and the *Residual Reduction* ($RR = 1 - |y^{(2)} - y^*| / |y^{(1)} - y^*|$), measuring whether Stage 2 reduces error. These metrics complement the unified framework by capturing adjustment dynamics unique to the self-generated paradigm.

Parse rate. The fraction of prompts for which an integer in $[0, 100]$ was successfully extracted from the model’s response, using a regex-first pipeline with an optional LLM fallback (§5).

5 Pilot validation

We validate the benchmark design with pilot studies on all five suites—external, history, ICL, RAG, and tool—using unified metrics (§4) that enable direct cross-suite comparison.

5.1 Pilot setup

External pilot. 90 items from the external suite (15 per domain, 6 domains), each rendered in 5 conditions (control, irrelevant_low, irrelevant_high, plausible_low, plausible_high), yielding 450 prompts.

History pilot. 60 items from the history suite (10 per domain, 6 domains), each rendered in 5 conditions using the two-stage protocol (§3.4), yielding 300 prompts. Non-control conditions require two inference stages per prompt: Stage 1 produces the model’s self-generated anchor, which is then retained in conversation history for Stage 2.

RAG pilot. 180 items from the RAG suite (30 per domain, 6 domains), each rendered in 5 conditions using the three-document corpus design (§3.4), yielding 900 prompts. All conditions share the same evidence and question; only the anchor-slot document varies.

ICL pilot. 180 items from the ICL suite (30 per domain, 6 domains), each rendered in 5 conditions using the metadata-priming design (§3.4), yielding 900 prompts. Three few-shot demonstrations precede the target item. All conditions share the same target evidence, question, demo evidence, and demo answers; only the demo header metadata varies (case IDs for irrelevant, prior estimates for plausible, plain headers for control).

Tool pilot. 180 items from the tool suite (30 per domain, 6 domains), each rendered in 5 conditions using the structured tool-output design (§3.4), yielding 900 prompts. All conditions share the same evidence and `get_evidence_summary` output; only the `check_external_reference` result varies. Prompts are rendered through each model’s native chat template with `role: "tool"` messages.

Models. Four instruction-tuned models spanning two families and two scale tiers: Qwen2.5-7B-Instruct, Qwen2.5-3B-Instruct (Team, 2024), Llama-3.1-8B-Instruct, and Llama-3.2-3B-Instruct (Grattafiori et al., 2024). All models are served on NVIDIA H100 GPUs with greedy decoding (temperature = 0.0, max tokens = 512).

Parsing. Integer answers are extracted via a regex parser that locates the last standalone integer in $[0, 100]$. When the regex fails, an LLM fallback (Qwen2.5-1.5B-Instruct) extracts the final numeric answer. Parse rates exceed 98% on the external, history, ICL, and RAG pilots for all models (the ICL pilot achieves 100% across all four models). On the tool pilot, Qwen models maintain high parse rates ($\geq 98\%$), while Llama models show reduced rates (79–86%), reflecting the additional challenge of generating well-formatted answers after structured tool-calling context.

Model	MAE _c ↓	Acc ₁₀ ↑	UAI _{irr}	UAI _{pls}	TAR _{irr}	TAR _{pls}	Disc _Δ ↑
Qwen2.5-7B	7.18	76.7%	0.05	0.27	0.18	0.64	0.22
Llama-3.1-8B	8.97	75.3%	−0.22	0.43	0.32	0.74	0.66
Llama-3.2-3B	10.54	66.7%	0.23	0.43	0.46	0.67	0.19
Qwen2.5-3B	11.81	56.7%	0.05	0.30	0.27	0.54	0.25

Table 3: External-pilot results (90 items, 5 conditions per item). MAE_c and Acc₁₀ are computed on control-condition responses only. UAI: Unified Anchor Influence; TAR: Toward-Anchor Rate; Disc_Δ: UAI_{pls} − UAI_{irr}. $\varepsilon = 3$ for UAI filtering.

Model	MAE _c ↓	Acc ₁₀ ↑	UAI _{irr}	UAI _{pls}	TAR _{irr}	TAR _{pls}	Disc _Δ ↑
Qwen2.5-7B	6.92	85.0%	−0.08	0.35	0.41	0.57	0.43
Llama-3.1-8B	7.97	81.7%	0.15	0.17	0.37	0.45	0.02
Llama-3.2-3B	9.88	73.3%	0.05	0.54	0.53	0.65	0.49
Qwen2.5-3B	11.70	50.0%	−0.17	0.25	0.42	0.58	0.41

Table 4: History-pilot results (60 items, two-stage protocol). MAE_c and Acc₁₀ are computed on control-condition responses only. Anchor values are model-generated (Stage 1 answers). ACR and RR are reported in Appendix B.

5.2 External pilot results

Table 3 reports the unified metrics for the external suite.

Accuracy and discrimination are different dimensions. Qwen2.5-7B (MAE 7.18, Acc 76.7%) and Llama-3.1-8B (MAE 8.97, Acc 75.3%) have comparable task accuracy, yet their anchoring profiles differ qualitatively. Llama-3.1-8B achieves the highest discrimination gap (Disc_Δ = 0.66), driven by a mild contrast effect under irrelevant anchoring (UAI_{irr} = −0.22) combined with substantial plausible sensitivity (UAI_{pls} = 0.43, TAR = 0.74). Qwen2.5-7B shows low anchoring overall (UAI_{irr} = 0.05, UAI_{pls} = 0.27) with moderate discrimination (0.22).

TAR reveals directional effects that UAI magnitude can miss. Llama-3.2-3B shows substantial irrelevant influence (UAI_{irr} = 0.23, TAR_{irr} = 0.46) and plausible influence (UAI_{pls} = 0.43, TAR_{pls} = 0.67), yielding low discrimination (Disc_Δ = 0.19). This indiscriminate anchoring pattern—high sensitivity regardless of relevance—is invisible to accuracy metrics alone: Llama-3.2-3B achieves better accuracy (66.7%) than Qwen2.5-3B (56.7%).

5.3 History pilot results

Table 4 reports the unified metrics for the history suite. For this suite, the anchor value is the model’s own Stage 1 answer rather than an experimenter-defined value (§4).

Self-generated anchoring patterns differ from external anchoring. Under the unified framework, the History suite reveals anchoring patterns that diverge sharply from the External suite. Llama-3.1-8B, which achieves the highest External-suite discrimination (Disc_Δ = 0.66), shows near-zero discrimination on History (Disc_Δ = 0.02): its UAI_{irr} (0.15) and UAI_{pls} (0.17) are nearly equal, indicating that it responds similarly to its own prior answers regardless of whether they pertain to the same or a different case.

Smaller models show higher plausible sensitivity to self-generated anchors. Llama-3.2-3B achieves the highest UAI_{pls} across all models and suites (0.54, TAR_{pls} = 0.65), indicating that its Stage 2 answers shift substantially toward its biased Stage 1 estimates—consistent with the insufficient adjustment pattern predicted by Epley & Gilovich (2001). Combined with low irrelevant influence (UAI_{irr} = 0.05), this yields the highest History-suite discrimination (Disc_Δ = 0.49).

Model	MAE _c ↓	Acc ₁₀ ↑	UAI _{irr}	UAI _{pls}	TAR _{irr}	TAR _{pls}	Disc _Δ ↑
Llama-3.1-8B	6.58	80.0%	0.07	0.24	0.34	0.51	0.17
Qwen2.5-3B	9.65	64.4%	0.00	0.10	0.18	0.38	0.10
Llama-3.2-3B	9.97	67.2%	0.11	0.07	0.41	0.44	−0.04
Qwen2.5-7B	11.77	50.6%	−0.01	0.22	0.13	0.51	0.23

Table 5: RAG-pilot results (180 items, 5 conditions per item). MAE_c and Acc₁₀ are computed on control-condition responses only.

Model	MAE _c ↓	Acc ₁₀ ↑	UAI _{irr}	UAI _{pls}	TAR _{irr}	TAR _{pls}	Disc _Δ ↑
Qwen2.5-7B	0.54	98.9%	0.00	0.16	0.11	0.44	0.16
Qwen2.5-3B	3.45	92.6%	0.06	0.09	0.24	0.29	0.04
Llama-3.1-8B	15.04	56.3%	0.13	0.51	0.28	0.65	0.38
Llama-3.2-3B	18.84	35.6%	0.22	0.39	0.38	0.55	0.17

Table 6: Tool-pilot results (180 items, 5 conditions per item). Anchors appear in structured **role: "tool"** messages rendered through each model’s chat template. MAE_c and Acc₁₀ are computed on control-condition responses only. Parse rates: Qwen2.5-7B 100%, Qwen2.5-3B 98.1%, Llama-3.1-8B 79.3%, Llama-3.2-3B 86.1%.

5.4 RAG pilot results

Table 5 reports the unified metrics for the RAG suite.

Document context reverses the accuracy ranking. On the External suite, Qwen2.5-7B achieves the best task accuracy (MAE 7.18); on the RAG suite, it is the worst (MAE 11.77, Acc 50.6%). Conversely, Llama-3.1-8B improves from MAE 8.97 (External) to MAE 6.58 (RAG), achieving the highest accuracy. This reversal indicates that evidence aggregation in the presence of retrieved documents is a distinct competence from estimation in prompt-only settings.

RAG context attenuates anchoring overall but does not eliminate it. Across all models, UAI_{pls} on RAG (range 0.07–0.24) is substantially lower than on External (range 0.27–0.43). The document context—which provides domain-relevant background alongside the anchor—appears to dilute anchor influence. However, Llama-3.2-3B shows negative discrimination on RAG (Disc_Δ = −0.04): it responds more to irrelevant document metadata (UAI_{irr} = 0.11, TAR = 0.41) than to plausible survey results (UAI_{pls} = 0.07, TAR = 0.44)—a concerning pattern for deployed RAG systems.

5.5 Tool pilot results

Table 6 reports the unified metrics for the tool suite, in which anchors appear within structured JSON tool outputs rendered via the model’s native chat template.

Structured tool context creates a sharp model-family divide. The tool suite reveals the starkest accuracy split of any suite: Qwen models achieve near-perfect evidence aggregation (MAE 0.54–3.45, Acc 93–99%), while Llama models struggle substantially (MAE 15–19, Acc 36–56%). This gap—absent on the external and history suites—suggests that the ability to extract and aggregate numeric evidence from structured **role: "tool"** messages is a distinct competence that varies across model families. The lower parse rates for Llama models (79–86% vs. 98–100% for Qwen) indicate that the tool-calling chat template itself poses a formatting challenge.

Anchoring susceptibility and tool-context competence co-vary across model families. Llama-3.1-8B shows the highest Tool-suite anchoring sensitivity (UAI_{pls} = 0.51, TAR_{pls} = 0.65) and the highest discrimination gap (Disc_Δ = 0.38). In contrast, Qwen2.5-7B nearly ignores irrelevant tool metadata (UAI_{irr} = 0.00) while remaining moderately sensitive to

Model	MAE _c ↓	Acc ₁₀ ↑	UAI _{irr}	UAI _{pls}	TAR _{irr}	TAR _{pls}	Disc _Δ ↑
Qwen2.5-7B	5.79	83.3%	−0.02	−0.01	0.19	0.28	0.00
Llama-3.1-8B	7.44	80.0%	0.14	0.19	0.48	0.52	0.05
Qwen2.5-3B	7.56	74.4%	−0.01	−0.04	0.14	0.19	−0.03
Llama-3.2-3B	6.61	82.2%	0.04	0.09	0.40	0.48	0.05

Table 7: ICL-pilot results (180 items, 5 conditions per item). Anchors appear only in demo-header metadata; demo evidence and answers are identical across conditions. MAE_c and Acc₁₀ are computed on control-condition responses only. Parse rate is 100% for all models.

450 plausible reference values (UAI_{pls} = 0.16)—a desirable anchoring profile for deployed agen-
 451 tic systems. Qwen2.5-3B shows near-zero discrimination (Disc_Δ = 0.04), indicating that it
 452 responds similarly to irrelevant and plausible tool outputs—not because it ignores both, but
 453 because its low overall anchoring sensitivity leaves little room for differentiation. However,
 454 the large accuracy and parse-rate gap between families means that anchoring susceptibility
 455 and structured-context competence are confounded in this suite: models that struggle to
 456 extract evidence from tool outputs may also be more susceptible to anchor values embedded
 457 in those outputs, and the present design cannot fully disentangle these factors.

458 5.6 ICL pilot results

459 Table 7 reports the unified metrics for the ICL suite, in which anchors appear only in the
 460 metadata headers of few-shot demonstrations—case IDs and batch numbers for irrelevant
 461 conditions, prior estimates for plausible conditions—while demo evidence and answers re-
 462 main identical across all conditions.

463 **Metadata priming produces the weakest anchoring of any suite.** Across all five suites, the
 464 ICL metadata-priming design yields the smallest UAI magnitudes. Both Qwen models show
 465 near-zero or slightly negative UAI (Qwen2.5-7B: UAI_{irr} = −0.02, UAI_{pls} = −0.01; Qwen2.5-
 466 3B: UAI_{irr} = −0.01, UAI_{pls} = −0.04), indicating effective immunity to incidental metadata
 467 numbers in demonstration context. Llama models show mild but detectable priming (Llama-
 468 3.1-8B: UAI_{pls} = 0.19, TAR_{pls} = 0.52), consistent with the selective accessibility mechanism:
 469 metadata numbers activate anchor-consistent retrieval that subtly influences the estimate.

470 **TAR confirms the absence of reliable toward-anchor shifts.** TAR values reinforce the con-
 471 clusion that metadata priming produces at most fragile effects. No model achieves TAR_{irr}
 472 above the 0.50 chance level (Llama-3.1-8B: 0.48; Llama-3.2-3B: 0.40; Qwen models: 0.14–
 473 0.19), indicating no systematic toward-anchor shift under irrelevant metadata. Only Llama-
 474 3.1-8B shows TAR_{pls} marginally above chance (0.52); the remaining models remain at or
 475 below 0.50. The family-level pattern is nevertheless informative: Qwen models show TAR
 476 well below chance on both relevance levels, consistent with effective immunity to metadata
 477 priming, while Llama models show near-chance TAR, indicating at most borderline sensi-
 478 tivity that does not reliably exceed chance.

479 5.7 Cross-suite comparison

480 Table 8 summarizes the key unified metrics across all five suites.

481 **Anchoring susceptibility is paradigm-dependent.** The five pilots collectively demonstrate
 482 that anchoring is not a unitary property of a model but depends on the delivery mechanism.
 483 Llama-3.1-8B exhibits the highest External-suite discrimination (Disc_Δ = 0.66) but near-
 484 zero History-suite discrimination (0.02), moderate RAG and Tool-suite discrimination (0.17
 485 and 0.38), and small ICL-suite discrimination (0.05). Llama-3.2-3B achieves the highest
 486 History-suite discrimination (0.49) but the lowest External-suite discrimination (0.19) and
 487 negative RAG-suite discrimination (−0.04). No model is consistently best or worst across
 488 paradigms.

Suite	Model	MAE _c ↓	UAI _{irr}	UAI _{pls}	Disc _Δ ↑
External	Qwen2.5-7B	7.18	0.05	0.27	0.22
	Llama-3.1-8B	8.97	−0.22	0.43	0.66
	Llama-3.2-3B	10.54	0.23	0.43	0.19
	Qwen2.5-3B	11.81	0.05	0.30	0.25
History	Qwen2.5-7B	6.92	−0.08	0.35	0.43
	Llama-3.1-8B	7.97	0.15	0.17	0.02
	Llama-3.2-3B	9.88	0.05	0.54	0.49
	Qwen2.5-3B	11.70	−0.17	0.25	0.41
ICL	Qwen2.5-7B	5.79	−0.02	−0.01	0.00
	Llama-3.2-3B	6.61	0.04	0.09	0.05
	Llama-3.1-8B	7.44	0.14	0.19	0.05
	Qwen2.5-3B	7.56	−0.01	−0.04	−0.03
RAG	Llama-3.1-8B	6.58	0.07	0.24	0.17
	Qwen2.5-3B	9.65	0.00	0.10	0.10
	Llama-3.2-3B	9.97	0.11	0.07	−0.04
	Qwen2.5-7B	11.77	−0.01	0.22	0.23
Tool	Qwen2.5-7B	0.54	0.00	0.16	0.16
	Qwen2.5-3B	3.45	0.06	0.09	0.04
	Llama-3.1-8B	15.04	0.13	0.51	0.38
	Llama-3.2-3B	18.84	0.22	0.39	0.17

Table 8: Cross-suite comparison using unified metrics. Models are ordered by MAE_c within each suite. The same model can rank differently across suites on both accuracy and discrimination.

The accuracy ranking reverses across suites. The model ranking by task accuracy is consistent across External and History suites (Qwen2.5-7B > Llama-3.1-8B > Llama-3.2-3B > Qwen2.5-3B) but reverses on the RAG suite, where Llama-3.1-8B leads and Qwen2.5-7B trails. The Tool suite introduces a third ranking pattern: Qwen models dominate (MAE 0.54–3.45) while Llama models lag substantially (MAE 15–19). The ICL suite introduces yet another ranking: Qwen2.5-7B leads (MAE 5.79) while Llama-3.2-3B outperforms both Llama-3.1-8B and Qwen2.5-3B, suggesting that evidence aggregation competence is context-format-dependent.

Plausible anchoring is consistently stronger than irrelevant anchoring. Across all 20 model–suite combinations, UAI_{pls} exceeds UAI_{irr} in 18 of 20 cases (Disc_Δ > 0), with two exceptions: Llama-3.2-3B on RAG (−0.04) and Qwen2.5-3B on ICL (−0.03). This overall pattern is encouraging—models generally *do* distinguish relevance levels—but the magnitude and consistency of discrimination vary substantially across paradigms, confirming the value of multi-suite evaluation.

Metadata priming is the subtlest anchoring pathway. The ICL suite yields the smallest anchoring effects of any suite: UAI_{pls} ranges from −0.04 to 0.19, compared to 0.07–0.54 across the other four suites. This is consistent with the selective accessibility theory: incidental metadata numbers in demonstration headers are a weaker prime than explicit anchor sentences (External), self-generated estimates (History), document content (RAG), or structured tool outputs (Tool). Qwen models are effectively immune to metadata priming (UAI near zero, TAR well below chance), while Llama models show near-chance TAR values (0.40–0.48 for irrelevant), indicating at most borderline directional sensitivity that does not reliably exceed the 0.50 chance level.

Structured tool outputs amplify model-family differences. The Tool suite’s accuracy gap between model families (Qwen MAE 0.5–3.5 vs. Llama MAE 15–19) far exceeds the spread observed on any other suite, where all four models cluster within a 5–8-point MAE range. At the same time, Tool-suite *anchoring* mirrors the External suite more closely than it does the RAG or ICL suites: Llama-3.1-8B achieves high discrimination on both External (0.66)

and Tool (0.38), while Qwen2.5-3B shows low discrimination on both (0.25 and 0.04). This partial correlation suggests that external and tool-output anchoring may share a susceptibility pathway, while document-based and metadata-priming anchoring engage different processing modes—though the confound between anchoring susceptibility and tool-context competence (§5.5) warrants caution in interpreting the Tool-suite patterns.

6 Limitations and future work

Suite coverage. The current experimental validation covers all five suites (External, History, ICL, RAG, and Tool) across four models. The five-suite pilots demonstrate rich paradigm-dependent anchoring patterns—including a model-family accuracy split unique to tool outputs, cross-suite discrimination reversals, and the ICL suite’s confirmation that metadata priming is the subtlest anchoring pathway.

History suite: control structure. The History suite uses a single-stage control but a two-stage protocol for anchored conditions. This structural asymmetry means that observed differences could partly reflect the multi-turn format rather than anchoring alone. A two-stage control (with neutral Stage 1 content) would address this confound but would require careful design to avoid introducing its own priming effects.

ICL suite design. The ICL suite embeds irrelevant metadata numbers in demonstration context while keeping demo answers neutral. An alternative design using biased demo answers would test ICL distribution-matching rather than clean numeric priming; we chose the metadata-number design to isolate the priming mechanism, but acknowledge that the biased-answer variant may capture a complementary failure mode.

Tool suite: confounds and ecological validity. The primary Tool-suite evaluation uses injected tool outputs (the full tool-call exchange is pre-constructed in the prompt context). An ecological validation subset confirms that results generalize to genuine multi-turn function-calling interaction, but the ecological subset is small (~50 items, control + plausible_high only) and serves as a sanity check rather than a comprehensive evaluation. Additionally, the large accuracy and parse-rate gap between model families on the Tool suite means that anchoring susceptibility and structured-context competence are confounded: models that struggle with tool-output formatting may also show inflated anchoring metrics. Disentangling these factors would require models with comparable tool-context competence or explicit controls for formatting ability.

Model coverage. The pilot evaluates four open-weight models from two families (Qwen, Llama) at 3B and 7–8B scale. Larger models, API-served models (GPT, Claude, Gemini), and models with explicit reasoning capabilities remain untested. We do not claim that the patterns observed in the pilot generalize across all model families and scales.

Single language. All items are in English. Anchoring susceptibility may vary across languages, particularly for models with different language training distributions.

Synthetic anchors. Benchmark anchors are constructed rather than naturally occurring. Real-world anchors encountered in deployed systems may differ in salience and framing.

Future directions. The benchmark is designed to support several extensions beyond its current scope:

- **Extended-scale evaluation** across all five suites with larger item counts and the difficulty axis, enabling finer-grained cross-paradigm comparisons. The current pilot already reveals that anchoring susceptibility is paradigm-dependent; larger samples will enable per-domain and per-difficulty analyses.
- **Mitigation evaluation:** the benchmark can serve as a controlled testbed for inference-time debiasing strategies, measuring whether mitigations reduce irrelevant anchoring without suppressing appropriate sensitivity to informative context.

- **Mechanistic analysis:** the paired-condition design (identical prompts differing only in the anchor component) supports activation patching and logit-lens analyses to localize where in the transformer architecture anchoring arises. The History suite’s two-stage protocol additionally enables comparison of internal representations between Stage 1 and Stage 2 to study how prior estimates persist across turns.
- **Informative-anchor validation:** extending the relevance axis to include genuinely informative anchors, testing whether models appropriately increase sensitivity when the anchor is evidentially justified.

Author contributions

[TODO: Fill in author contributions before submission.]

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Ethics statement

This work studies a failure mode of LLMs that could lead to systematically biased outputs in high-stakes applications. Our benchmark uses synthetic scenarios in health, finance, and operations domains; no real patient, financial, or operational data is used. We release the benchmark to enable further research on LLM robustness.

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634 A Experimental setup details

635 The experimental setup for all five pilots (external, history, ICL, RAG, and tool) is described
636 in §5.1. This section provides additional details on infrastructure and reproducibility.

637 **Inference.** Open-weight models are served via HuggingFace Transformers (bfloat16,
638 `device_map=auto`) on NVIDIA H100 GPUs (95 GB VRAM). All experiments use greedy
639 decoding (temperature = 0.0) with max tokens = 512 and a fixed random seed (42). For
640 the History suite’s two-stage conditions, Stage 1 and Stage 2 are executed sequentially within
641 a single conversation using the tokenizer’s chat template. For the Tool suite, prompts are
642 rendered through each model’s native tool-calling chat template, with `role: "tool"` mes-
643 sages conveying structured JSON tool outputs. For the ICL suite, prompts are plaintext
644 with three few-shot demonstrations; no chat template or tool messages are required.

645 **Parsing pipeline.** We employ a two-stage parsing pipeline: (1) a regular expression extracts
646 the last standalone integer in $[0, 100]$ from the response; (2) if the regex fails, an LLM fallback
647 (Qwen2.5-1.5B-Instruct) extracts the final numeric answer. All models achieve $\geq 98\%$ parse
648 rate on the external, history, ICL, and RAG pilots (the ICL pilot achieves 100% across all
649 four models). On the tool pilot, Qwen models maintain $\geq 98\%$ parse rates, while Llama
650 models achieve 79–86%, reflecting the additional formatting challenge of structured tool-
651 calling context.

Parallelism. With four available H100 GPUs, all four models are evaluated in parallel (one model per GPU). The external pilot (450 prompts per model) completes in under 30 minutes per model; the history pilot (300 prompts, with two-stage inference for 240 of them) completes in approximately 1–21 minutes per model, depending on model size. The ICL, RAG, and tool pilots use batched inference (batch size = 32) for improved GPU utilization; 900 prompts per model complete in under 15 minutes per model.

Reproducibility. All five benchmark datasets are fully deterministic (seed = 42). The generation pipeline, evaluation code, and experimental artifacts are available at [URL redacted for review].

B History-suite secondary metrics

The History suite’s two-stage protocol produces a genuine self-generated anchor (the model’s Stage 1 answer) against which adjustment can be measured directly. Table 9 reports two adjustment-specific metrics that complement the unified UAI framework.

Model	ACR $\uparrow$	RR $\uparrow$
Qwen2.5-7B	0.85	0.34
Llama-3.1-8B	1.09	0.49
Qwen2.5-3B	0.42	−0.51
Llama-3.2-3B	0.29	−0.37

Table 9: History-suite secondary metrics (plausible conditions only). ACR: Adjustment Completion Ratio = $(y^{(2)} - y^{(1)}) / (y_{\text{evidence}}^* - y^{(1)})$; ACR = 1 is complete adjustment. RR: Residual Reduction = $1 - |y^{(2)} - y^*| / |y^{(1)} - y^*|$; RR > 0 indicates error reduction. Items with $|y^* - y^{(1)}| < 1$ are excluded.

Larger models adjust more completely. The 7–8B models achieve ACR values of 0.85 (Qwen2.5-7B) and 1.09 (Llama-3.1-8B), indicating near-complete or slight over-adjustment from Stage 1 to Stage 2. The 3B models achieve substantially lower ACR values (0.42 and 0.29), retaining roughly 60–70% of the Stage 1 bias after receiving full evidence. This scale-dependent pattern is consistent with the human anchoring literature, which finds that adjustment completeness increases with cognitive resources (Epley & Gilovich, 2001).

Residual reduction separates genuine correction from error persistence. Positive RR for the 7–8B models (0.49, 0.34) indicates that Stage 2 reduces the error introduced by biased Stage 1 evidence. Negative RR for the 3B models (−0.37, −0.51) indicates that these models fail to improve—and in fact amplify error—after seeing full evidence. This pattern complements the unified UAI framework: high UAI_{pls} on History (main table) indicates overall sensitivity to self-generated anchors, while low ACR and negative RR confirm that the sensitivity reflects genuine insufficient adjustment rather than a different mechanism.

C Mitigation strategies

The ANCHORBENCH design supports evaluation of inference-time mitigation strategies as a downstream use case. While mitigation is not the primary contribution of this work, we briefly describe six strategies implemented in the evaluation framework, organized into prompt-level and multi-pass categories.

Prompt-level (single-pass). **B1** (Ignore-Numbers): Prepends an instruction to ignore contextual numbers. **B2** (Bias-Warning): Prepends a warning about anchoring bias. **B3** (Chain-of-Thought): Requests step-by-step reasoning before answering.

686 **Multi-pass. SELFHELP** (Self-Debiasing): The model rewrites the prompt to remove bias-
 687 ing numbers, then answers the rewritten prompt. **LATE_BINDING** (Anchor-Late Reintro-
 688 duction): Pass 1 replaces the anchor with [REDACTED]; Pass 2 reveals the anchor and asks
 689 the model to update only if justified.

690 **Baseline. B0** (No Mitigation): The unmodified anchored prompt.

691 **Preliminary observations.** In prior experiments on the v1 benchmark (which used only the
 692 plausible relevance level), we observed that B2 (Bias-Warning) provided the most consistent
 693 anchoring reduction across models, while B3 (Chain-of-Thought) paradoxically amplified
 694 anchoring. Full mitigation evaluation on the v2 benchmark with the relevance axis is planned
 695 as future work and will measure whether mitigations reduce irrelevant anchoring without
 696 suppressing appropriate sensitivity to informative context.

697 D Mechanistic analysis

698 The paired-condition design of ANCHORBENCH (identical prompts differing only in the
 699 anchor-bearing component) is well-suited for mechanistic interpretability analyses such as
 700 logit-lens probing and activation patching. While mechanistic analysis is not the primary
 701 contribution of this benchmark paper, we briefly report preliminary findings from a v1-
 702 benchmark study to illustrate the benchmark’s utility for this purpose.

703 **Setup.** Using 200 stratified items from the v1 benchmark (which used the plausible rele-
 704 vance level only), we performed logit-lens and residual-stream patching analyses on three
 705 open-weight models: Llama-3.2-1B, Llama-3.2-3B, and Qwen3-4B-FP8.

706 **Key observations.** (1) The anchor-control JS divergence remains near zero in early trans-
 707 former layers and rises sharply in the final third of the network, indicating that anchoring
 708 distortion is introduced or amplified in late layers. (2) Attention-head attribution reveals
 709 architecture-dependent patterns: Qwen3-4B shows concentrated anchoring effects in a small
 710 number of heads (head 0 in layers 20–27, effect sizes 0.40–0.46), while the Llama models
 711 show distributed attribution with smaller individual effects (0.06–0.07). (3) These findings
 712 are preliminary and based on models at 1B–4B scale; generalization to larger models and
 713 other architectures is an open question.

714 **Benchmark utility.** These results demonstrate that ANCHORBENCH’s paired-condition de-
 715 sign enables clean causal attribution of anchoring effects—a capability that, combined with
 716 the relevance axis, could support future investigations into whether models process irrelevant
 717 and plausible anchors through different internal pathways.

718 E Reproducibility

719 **Dataset.** The benchmark is generated by the pipeline in `src/anchorbench_v1/` and stored
 720 as JSONL files (one for ITEMSPECS, one for PROMPTVIEWS). Five pilot datasets are used
 721 in the reported experiments:

- 722 • **External pilot:** 90 items $\times$ 5 conditions = 450 prompts (15 per domain, 6 domains).
- 723 • **History pilot:** 60 items $\times$ 5 conditions = 300 prompts (10 per domain, 6 domains),
 724 with each non-control condition requiring two inference stages.
- 725 • **ICL pilot:** 180 items $\times$ 5 conditions = 900 prompts (30 per domain, 6 domains),
 726 with three few-shot demonstrations per item.
- 727 • **RAG pilot:** 180 items $\times$ 5 conditions = 900 prompts (30 per domain, 6 domains),
 728 with a frozen three-document corpus per item.
- 729 • **Tool pilot:** 180 items $\times$ 5 conditions = 900 prompts (30 per domain, 6 domains),
 730 with structured tool outputs rendered via each model’s chat template.

731 All generation is fully deterministic (seed = 42).

732 **Inference.** Open-weight models are served via HuggingFace Transformers (bfloat16,
733 `device_map=auto`) on NVIDIA H100 GPUs (95 GB VRAM). All experiments use greedy
734 decoding (temperature = 0.0, seed = 42, max tokens = 512). The History suite’s two-stage
735 inference uses the tokenizer’s `apply_chat_template` to construct multi-turn conversations
736 with the model’s own Stage 1 response as an assistant turn. The Tool suite renders prompts
737 through each model’s native tool-calling chat template with `role: "tool"` messages. The
738 ICL suite uses plaintext prompts with three few-shot demonstrations; no chat template or
739 tool messages are required.

740 **Evaluation.** Each suite has a dedicated evaluation script
741 (`scripts/eval/run_{suite}_v2.py`) that loads the corresponding pilot dataset, runs
742 inference, applies the two-stage parsing pipeline (regex $\rightarrow$ LLM fallback via Qwen2.5-
743 1.5B-Instruct), and computes the unified metrics (UAI, TAR, Disc_Δ , MAE, Acc_{10} ,
744 parse rate). The History-suite script additionally orchestrates two-stage inference
745 and computes ACR and RR. All evaluation scripts share a common metrics module
746 (`scripts/eval/unified_metrics.py`).

747 **Hardware.** All experiments run on a server with $4\times$ NVIDIA H100 GPUs (95 GB VRAM
748 each). Four models are evaluated in parallel (one per GPU). The external pilot completes
749 in under 30 minutes per model; the history pilot in 1–21 minutes depending on model size.
750 The ICL, RAG, and tool pilots use batched inference (batch size = 32) and complete in
751 under 15 minutes per model.

752 **Code and data availability.** The full codebase, benchmark generation pipeline, evaluation
753 scripts, and pilot datasets are available at [URL redacted for review].